

Methods: Instructions that dictate the type of action the client wishes to perform. Common hypertext transfer protocol (HTTP) methods include GET (to retrieve data), POST (to send data or create a resource), PUT (to update an existing resource), and DELETE (to remove a resource).

Parameters: Specific details required for a request, such as location data for a weather API or authentication credentials for accessing a secure service.

Responses: The data sent back by the server application in reply to a request, commonly formatted in JavaScript Object Notation (JSON) or Extensible Markup Language (XML).

Additional components supporting the API ecosystem include API designers, developer portals for discovering and understanding APIs, the API backend that executes the requested actions, and API gateways that manage incoming requests, apply usage rules (such as authentication and rate limiting), and route them accordingly.

APIs are not merely technical conveniences; they are fundamental enablers of interconnected digital ecosystems. The design, openness, and accessibility of these interfaces directly shape the diversity and inclusivity of these digital environments. If APIs are complex, poorly documented, or proprietary and expensive, they can inadvertently create barriers, limiting participation to only those with significant resources or technical expertise. Conversely, well-designed, open, and standardised APIs can foster broader participation, democratising access to digital tools and services. The widespread adoption of standardised API protocols, such as Representational State Transfer (REST) and GraphQL, alongside standard data formats like JSON, has been a crucial factor in their success. This standardisation simplifies the process of integrating disparate systems, reducing complexity and lowering the barrier to entry for developers and organisations of all sizes, including those in underserved

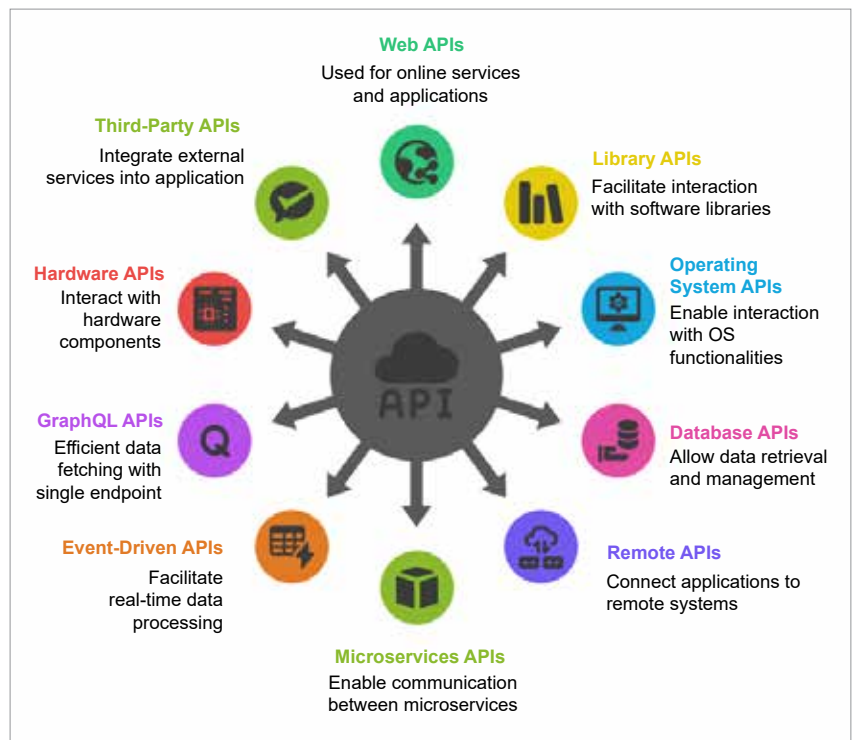


Figure1: Types of APIs and their applications

regions. This broader capacity for participation in the digital economy is a direct mechanism through which APIs can help bridge digital divides.

APIs come in various forms, each serving different purposes. Web APIs are the most common, allowing web applications to interact with each other over the internet. Examples include RESTful APIs and SOAP APIs. Library APIs provide functions and procedures for developers to interact with software libraries. Operating system APIs enable applications to interact with the system, allowing functionalities such as file management and process control. Database APIs enable applications to interact with databases, allowing for the retrieval, insertion, and management of data.

The real-world benefits

The numerous benefits of using APIs make them indispensable in modern software development. By leveraging existing functionalities, APIs save developers time and effort, allowing them

to focus on building unique features. APIs enable applications to scale by integrating with other services and systems, ensuring seamless performance even as user demand grows. They provide a controlled way to access data and functionalities, ensuring only authorised applications can interact with the system. APIs foster innovation by enabling developers to build upon existing services.

APIs act as bridges that connect disparate systems and unlock data and functionalities. They are ubiquitous in today's digital landscape, powering many applications and services. Social media integration is a prime example, where APIs enable applications to integrate with social media platforms, allowing users to share content and interact with their social networks. Payment processing is another critical application, where APIs facilitate secure and efficient payment processing, enabling e-commerce sites to accept payments from various providers. Weather APIs provide real-time weather information to applications, allowing users to access accurate forecasts and